

AI GUIDELINES

EXPLORING POINTS OF CONVERGENCE BETWEEN FAITH-BASED AND SECULAR STATEMENTS

Ezekiel Kwetchi Takam

In June 2019, the article “Artificial Intelligence: The Global Landscape of Ethics Guideline” was published.¹⁹ This influential article set out to analyse points of convergence among different ethical guidelines on artificial intelligence (AI) produced over the last five years. To accomplish this, the authors used a database of 84 ethical guidelines published by private companies, government agencies, academic and research institutions, intergovernmental organisations, science foundations, federations of worker unions and political parties. The study showed a global convergence around five ethical principles: transparency, justice and fairness, non-maleficence, responsibility, and privacy. These principles are also found in two faith-based ethical guidelines: *The Rome Call for AI Ethics* (hereafter referred to as the Rome Call), published by The Pontifical Academy of Life, and *Artificial Intelligence: An Evangelical Statement of Principles* (hereafter referred to as the Baptist Statement), published by the Southern Baptist Convention²⁰.

¹⁹ Jobin, Anne, Ienca, Marcello and Vayena, Effy, “Artificial Intelligence: Global Landscape of Ethics Guidelines,” *Nature Machine Intelligence*, 1(2019), 389-399. <https://arxiv.org/ftp/arxiv/papers/1906/1906.11668.pdf>

²⁰ “Rome Call for AI Ethics,” <https://www.romecall.org>; The Southern Baptist Ethics & Religious Liberty Commission (ERLC), 2019, “Artificial Intelligence:

In this chapter, we will highlight the five convergent principles in these two faith-based ethical guidelines. This exercise is relevant for two reasons. On the one hand, it shows through its comparative approach that there are substantial and unanimous ethical points in the existing ethical guidelines, whatever the nature of the producing entities (confessional or non-confessional). On the other hand, the above-mentioned converging principles, are necessary for the productivity of educational systems in the era of artificial intelligence. Consider the goals of access to education and lifelong learning. Artificial intelligence can indeed meet both objectives as an adaptive learning tool; (which analyzes the learner's traces and abilities, in order to propose personalised and adapted learning). However, in order to adapt to different learners, its design must include different learner profiles, taking into account different parameters of ages, cultures, geographical regions etc. In other words, this AI must be fair (second of the five convergent principles) in the sense of inclusion; Hence the importance of the principle of justice/fairness, alongside the other four converging principles. In our following remarks, we will present the different schools of meaning of these five principles and establish the link with the meanings developed in The Rome Call and the Baptist Statement.

Transparency

The concept of transparency is present in 73 of the 84 documents analysed and is often combined with the notions of explicability and interpretability. Its fields of application include the use of data, algorithmic decisions, and the relationship between humans and AI. The study structures the presentation of this transparency in two points: the

definition of the concept and the means necessary for its effectiveness (or its instrumental component).

Speaking of the definition, the authors of the study specify that it does vary across guidelines. Three main schools of thought emerge. The first defines transparency as a means of minimising the risks of AI and improving it; the second school emphasises its legal dimension and its role in building trust; the third school links it to the notions of dialogue, participation, and the principles of democracy.²¹

As regards the instrumental component, the study also identifies three schools. The first is that of auditing, which involves the democratisation and availability of information on the development and application of AI systems. According to the guidelines, this information to be democratised varies between lines of code and data typologies, without forgetting the limits of AI, its uses and its intentions. The second school, largely made up of the private sector, recommends a technical solution. This includes coding the algorithms to such an extent that they are transparent. The third school, which is like the first, recommends dialogue between developers and the public during the design process in order to build trust.

Transparency is the first principle of *algorithethics*. The approach to this notion, as proposed by the Rome Call, is in line with the third school of non-confessional guidelines in terms of definition (the school of dialogue, citizen participation and principles of democracy) as well as the second and third schools of the instrumental component. The latter encourage technical solutions (second school) and dialogue between developers and citizens (third school) respectively.

The Baptist Statement also deals with the principle of transparency. Article 8 of the declaration, among other recommendations, emphasises the concept of informed consent which must be preceded by a consistently good intention. The Baptist Statement is thus close to the

²¹ The Southern Baptist Ethics & Religious Liberty Commission, op. cit., 8.

second school of the non-denominational guidelines in terms of definition (building trust) and to the first school of the instrumental component (making information available for audit).

Responsibility

The principle of responsibility was addressed in 60 of the 84 guides analysed. These are structured into three main schools of thought which focus solely on the means necessary for the effectiveness of this principle (the instrumental component). The first proposes a legal reading of responsibility, which could be deployed at the beginning of the technological creation process through the approach of contracts, or at the end through the approach of recourse. The second school focuses on the process of creation and recommends a particular exploration of the underlying mechanisms that can lead to harmful actions. The third school emphasises the need for whistleblowing and redress in the event of danger and the promotion of diversity and education.

The *responsible* actors vary from AI developers, designers, institutions, and so forth. The study also notes that there is a debate about the possibility of attributing responsibility to AI.

In the Rome Call, responsibility is the third principle of algorethics and it is attributed to developers. The principle states that developers, in their creative process, must be transparent and accountable. Through this reading, the Pontifical guideline fits into the second school of non-denominational guidelines, as does the Baptist Declaration. The latter, in its Article 3, which deals with the relationship between AI and humanity, opposes the possibility of attributing any responsibility to AI. This position is reaffirmed in Article 10, which deals with war.²²

²² Ibid.

Privacy

Privacy is mentioned in 47 of the 84 guides analysed. In terms of definition, two schools of thought exist. The first associates privacy with the notions of data protection and security, while the second links it to the principles of freedom and trust.

As regards the means necessary for the effectiveness of this privacy, the authors of the study identify three schools. The first is that of technical solutions (privacy by design). The second school encourages more research and awareness-raising to make users aware of the issues at stake. The third school opts for the regulatory approach, which consists of regulating AI by creating specific laws and regulations.

Privacy is the sixth principle of algorethics in the Rome Call. It states that AI systems must operate safely and respect the privacy of users. The Rome Call is thus in the first school of non-confessional guidelines in terms of definition. Concerning the register of means necessary for its effectiveness, the call joins the first and second schools which insist respectively on technical solutions (privacy by design) and on education as a tool for raising awareness of the challenges of AI.

Article 8 of the Baptist Statement is introduced by presenting privacy and private property as inseparable from the rights and choices of each individual.²³ This position is in line with the first school of non-confessional guides in terms of definition and the third school in terms of the means necessary to implement the principle.

Non-maleficence

This principle is recommended by 60 of the 84 guides analysed. It is combined with the notions of security, safety, protection, precaution, prevention, and integrity (physical or mental). These guidelines are

²³ Ibid.

opposed to the development and use of AI that would undermine the integrity of humans, their psychological and emotional well-being.

According to the Rome Call “[A]ny AI-based technology must never be used to exploit people in any way, especially those who are most vulnerable.”²⁴

Article 4 of the Baptist Statement defines non-maleficence as one of the criteria to be met by AI in the service of medicine.²⁵ Article 11 on war opposes the use of AI for reasons of torture, genocide, and so forth.²⁶ Article 9 on security also opposes the use of AI that would undermine freedom of expression.²⁷

Justice and Fairness

Recommended by 68 of the 84 guides analysed, justice/fairness is associated with the notions of inclusion, equality, non-discrimination, equity, diversity, plurality, and bias. The authors identify three schools of definition. The first is the ethical guidelines that define justice as respect for diversity, inclusion, and equality. The second school defines it as the ability to challenge decisions about artificial intelligence, including the right to object. The third school defines it as common access to AI technologies and services.

The means necessary for the effectiveness of this justice are based on five points: 1) technical solutions (justice by design); 2) transparency and awareness-raising among citizens through the democratisation of information relating to the development and deployment of AI; 3) auditing; 4) the strengthening of legal and judicial mechanisms to

²⁴ “Rome Call for AI Ethics,” <https://www.romecall.org/>.

²⁵ Artificial Intelligence: An Evangelical Statement of Principles, op. cit.

²⁶ Ibid.

²⁷ Ibid.

oppose actions that would harm citizens; and 5) the promotion of interdisciplinarity and diversity in the innovation process.

The notion of justice comes close to the second principle of algoethics (Inclusion: the needs of all human beings must be taken into consideration so that everyone can benefit, and all individuals can be offered the best possible conditions to express themselves and develop) and to the fourth principle (Impartiality: do not create or act according to bias, thus safeguarding fairness and human dignity). In this reading, the Rome Call falls within the register of the three definitional schools mentioned above (inclusion; the importance and existence of legal mechanisms to counter unfair decisions and actions; and equal access to the benefits of AI). It is, however, intended to emphasise legal mechanisms for redress (the second school) and equal access to the benefits of AI (the third school).²⁸ In relation to the means necessary for the effectiveness of justice, the Rome Call is in line with the first, second and third schools of non-confessional guides. These recommend respectively technical solutions, transparency and awareness, and legal devices.

²⁸ In the first paragraph of the chapter on rights, the Rome Appeal argues that the “ethical commitment of all stakeholders involved [in the development of artificial intelligence] is a crucial starting point; in order to make this future a reality, values, principles and, in some cases, legal regulations, are absolutely essential to support, structure and guide this process.” The last paragraph of the same chapter states that “new forms of regulation must be encouraged to promote transparency and respect for ethical principles, especially for advanced technologies that present a higher risk of impacting on human rights, such as facial recognition. Speaking of common access to the benefits of AI (the third school of definition). The Rome Appeal supports the primacy of the disadvantaged and marginalised. It states that the “development of AI for the benefit of humanity and the planet must be reflected in regulations and principles that protect people - especially the weak and disadvantaged - and the natural environment.”

Article 4 of the Baptist Statement defines justice as one of the main criteria that artificial intelligences in the service of medicine will have to respect.²⁹ Article 5, which deals with algorithmic biases, insists that non-discrimination must characterise AI: “AI must be designed and used in a way that treats all human beings as having equal value and dignity.”³⁰ These articles are thus close to the first school of non-confessional guides in terms of definition (inclusion and equality). Talking about the means necessary for the effectiveness of this principle of justice, the Baptist Statement is in line with the second school of non-denominational guidelines, through its article 11 which stresses the need to involve citizens in defining the role of the AI.

In the end, these five principles are the most represented in the guidelines analysed, alongside other principles such as trust (present in 28 of the 84 ethical guidelines); solidarity (present in 6 of the 84); sustainability (present in 14 of the 84); dignity (13 of the 84); beneficence (41 of the 84); freedom and autonomy (present in 34 of the 84). While this convergence of principles shows to some extent that religious institutions have a place in the international and interdisciplinary debate on the ethics of AI, it also calls into question the plurality of existing ethical guidelines. If there are converging principles, regardless of the institutions that produce them, would it not be possible and preferable to think of a single ethical guide for AI? Moreover, aware that AI, reinforced by globalisation, is increasingly proving to be a technology with a universal impact, could these five key and converging ethical principles not form the basis of a single and universal ethical declaration of AI, similar to the Universal Declaration of Human Rights?

²⁹ Artificial Intelligence: An Evangelical Statement of Principles, op. cit.

³⁰ Ibid.

Conclusion

The objective of this work was to propose a brief analyse and comment of two ethical AI guidelines produced by religious institutions in contrast to similar non-confessional efforts. We have thus highlighted the five converging principles that are: transparency, justice and fairness, non-maleficence, responsibility and privacy. At the end of this exercise, important questions remain – questions that deserve further reflection. These include questions related to the definition of the relationship between human and AI, the eco-responsibility approach of AI, and the redefinition of education in the AI context.

The first set of questions, those pertaining to the definition of the relationship between human and AI, brings up two issues: the human responsibility and the possibility of considering AI as an electronic person with rights and duties. According to both the Rome Call and the Baptist Statement, the responsibility of the actions of AI must lie with the human creator. This defense of human responsibility is often associated with the denial of the legal recognition of robots as *electronic persons* with rights and duties. The above analysis shows that the human monopoly of responsibility is not contrary to the legal recognition of the rights and duties of AI. Indeed, the arguments against the legal recognition of AI-robots as *electronic persons* are based on two key notions that would be definitional substances of the *person*: free will and the corporal envelopment. However, several neuroscientific and biological studies have challenged these convictions.³¹ They invite us to

³¹ “Le libre arbitre existe-t-il ?” *Science étonnante*, 5 March 2012, <https://scienceetonnante.com/2012/03/05/le-libre-arbitre-existe-t-il/> (accessed 3 September 2020); Campa, Riccardo, Corbally, Christoffer and Rappaport, Margaret. “Electronic Persons. Is it Premature to Grant Personhood to Machines but Never Say Never,” https://www.researchgate.net/profile/Riccardo-Campa/publication/348592771_Electronic_persons_It_is_premature_to_grant_personhood_to_machines_but_never_say_never/links/6006bd56299b14088a64d

redefine the concept of responsibility by dissociating it from the problem of the rights and duties of robots. Responsibility must be attributed to human, not because we have free will or a body that would make us a *person*, but for the simple reason that we are the creator and that it is our neural mechanisms that are transposed and reproduced in AI. It is therefore a part of the human, digitised or algorithmised, that is at work in such AI. In relation to these two questions, several theological avenues (not taken up in the two statements) come to mind, notably the concept of *Imago Dei*. On the one hand, the concept of *Imago Dei* affirms the particularity and dignity of humans who are in the image of God, while at the same time attesting to their freedom and equality. It is therefore, from this perspective, a basic theology for the ethical triptych of AI, which is: the protection of freedom, the guarantee of equality and the central importance of human dignity. On the other hand, this concept defines human creativity as a reflection of divine creativity. In doing so, AI, which is part of human creativity, is a counterpart of the human that helps the latter to better understand ourselves (functionalist understanding) and our relationship to God.³²

As regard to the second set of questions relating to the eco-responsible approach of AI, the Rome Call insists that the environment and its wellbeing must be a central concern in the research, the development and the application of AI. In relation to this position, one can think of an ecotheological approach to AI as a theological extension. This can be articulated in the light of the concept of holism, which was developed at length by the theologian Thomas Berry. According to holism, the human from birth to adulthood is a being that is required to

1f/Electronic-persons-It-is-premature-to-grant-personhood-to-machines-but-never-say-never.pdf

³² This concept is widely used by theologian Anne Foerst in her article: Foerst, Anne. "Cog, a Humanoid Robot, and the Question of the Image of God". *Zygon*®, 1998, 33: 91-111. <https://doi.org/10.1111/0591-2385.1291998129>

coexist inextricably with the *other* to survive. This *other* includes both human and non-human entities (plant, animal, and so on). The ecotheological approach therefore consists in sacralising this whole, this relationship, this interdependence, by making the history of humans, the history of God, and the history of the earth *one*.³³ It is in this vein of theological reflection that we can see the importance of the other. It is in this vein of ideas that the ecotheologian Sallie McFague proposes to redefine the human way of understanding God's presence in the world. In response to the ecological crisis, she recommends seeing the world as the body of God.³⁴ This metaphor allows for the sacredness of natural resources, the connectivity between all forms of earthly life, and more.

This also points to the direct need for a redefinition of the educational system (third set of questions), a core concern taken up in this collection of essays. Education was approached by the Rome Call from two angles. On the one hand, as a field of application of AI in order to guarantee, among other things, the principle of life-long learning; on the other hand, as a tool that makes humans distant and critical towards AI and its issues. On the other hand, one could also consider education as a tool that allows humans to communicate with AI, to learn from them, and thus to keep their critical spirit and freedom of thought. Hence the idea of redefining basic education from reading and writing to reading, writing, and coding. This is an initiative that is already being implemented in several countries, notably Japan, which since 2020 has made programming compulsory.

Such education in AI should also emphasise the particularities of the human, the values that define humanity such as courage, sympathy,

³³ For more, see Berry, Thomas, Swimme, Brian. *The Universe Story: From the Primordial Flaring Forth to the Ecozoic Era, a Celebration of the Unfolding of the Cosmos*, San Francisco: Harper, 1992.

³⁴ See McFague, Sallie. *The Body of God: An Ecological Theology* Minneapolis: Fortress Press, 1993.

benevolence, and, most importantly, the vital importance of peaceful relationship with the environment.

The guidelines studied above represent an important step in understanding how we understand the implicit rules of our common lives with AI. This work, however, is not static and requires ongoing debate and resources from all corners of society. As we converge together on an understanding of AI, the pedagogical, ethical element must not be lost by those who will craft the rules and implement them.